Name_____ Date_____ Period_____

LESSON 8-3 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Multiplication Equations

Maryland Standard 6.EE.B.7 · Learning target – I can solve a multiplication equation by dividing both sides by the same number.

REMEMBER

- In $6x$, the **coefficient** 6 means $6 \times x$.
- To solve $ax = b$, **divide both sides by** a. Division undoes multiplication.
- **Check**: the coefficient times your solution should equal the right side.

WORKED EXAMPLE

Solve $8p = 96$.



Step 1 Divide both sides by 8: $p = 96 \div 8 = 12$

Step 2 Check: $8 \times 12 = 96$, so it checks.

Answer $p = 12$

WORD BANK

coefficient – the number times the variable

divide both sides – undo the times

equal – the same amount

PRACTICE

show your work · label units

1. Solve each equation.

a. $4x = 36$ ____

b. $7n = 56$ ____

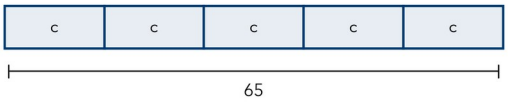
c. $9y = 108$ ____

d. $12k = 84$ ____

e. $5m = 85$ ____

f. $3w = 4.5$ ____

2. Write a multiplication equation for the tape diagram. Then solve it.



Equation and solution_____

3. Bus Fare. A family pays $11.25 for 5 bus fares. Each fare costs the **same**. Write and solve an equation for *f*, the cost of one fare. Check your answer.

Step 1 5 × one fare = total.

Step 2 Divide both sides by 5.

Show your work.

Answer_____ dollars

4. Garden Rows. Ms. Bell plants 84 seedlings in 6 equal rows. Write and solve a multiplication equation for *s*, the seedlings in each row.

Step 1 Rows × seedlings in each row = all seedlings.

Step 2 Divide both sides by 6.

Answer_____ seedlings

5. Compare. Without solving, tell which equation has the greater solution: $4n = 80$ or $5n = 80$. Explain. Then solve both to check.

Frame *___ has the greater solution because the same total split into ___ groups makes each group ___.*

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Find a package with equal parts (eggs, juice boxes, crayons). Write a multiplication equation for it and solve for one part.

Name_____ Date_____ Period_____

LESSON 8-3 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Division Equations

Maryland Standard 6.EE.B.7 · Learning target – I can solve a division equation by multiplying both sides by the same number.

REMEMBER

- To solve $x \div a = b$, **multiply both sides by** a.
- The solution is the **whole**, so it is greater than b.
- **Check**: divide your solution by a.

WORKED EXAMPLE

Solve $n \div 4 = 15$.

Step 1 Multiply both sides by 4: $n = 15 \times 4 = 60$

Step 2 Check: $60 \div 4 = 15$, so it checks.

Answer $n = 60$

WORD BANK

divisor – the number you divide by

multiply both sides – undo the divide

share equally – split into equal groups

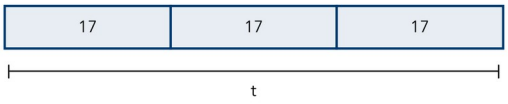
PRACTICE

show your work · label units

1. Solve each equation.

a. $n \div 7 = 6$ ____ **b.** $y \div 5 = 13$ ____ **c.** $k \div 10 = 4.5$ ____

2. Write a division equation for the tape diagram. Then solve it.



Equation and solution_____

3. **Jump Rope.** A rope is cut into 8 **equal** pieces. Each piece is 2.5 meters long. Write and solve a division equation for r, the whole rope.

Step 1 Whole rope ÷ 8 = one piece.

Step 2 Multiply both sides by 8.

Show your work.

Answer_____ meters

4. Pizza Party. A pizza is cut into equal slices. 6 friends share it, and each gets 2 slices. Write and solve a division equation for z, the slices in the pizza.

Step 1 Slices ÷ friends = slices for each friend.

Step 2 Multiply both sides by 6.

Answer_____ slices

5. Find the Error. Kayla and 3 friends share cards equally. Each gets 9 cards. Kayla writes $c \div 3 = 9$. Explain her mistake. Write and solve a correct equation.

Frame *Kayla's mistake is ___. The cards are shared by ___ people, so c = ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. Solve $y \div 6 = 9$.

A $y = 1.5$ **B** $y = 3$ **C** $y = 15$ **D** $y = 54$

7. 12 students share p pretzels equally. Each gets 5. Which equation fits?

A $p \div 12 = 5$ **B** $12 \div p = 5$ **C** $5p = 12$ **D** $p - 12 = 5$

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___